

Gibanje

- opis - kinematika
- napoved - dinamika

Kaj opisujemo? (od preprostih do zapletenih)

- točkasto telo (odvisno od tega, kakšno namučimo - npr. da Keplerjeve zakone je lahko Zemlja točk. telo)
- sistem točk. teles
- togo telo
- deformacije

Opazovalni sistem (kje in kdaj)

- koordinatni sistem
- ura
- za točk. telo:
 - radij vektor
 - $(\vec{r} = (x, y, z) \text{ [m]})$

+ Opazujemo pujsa na toboganu

- če se pujs kotli, ne more bit točk. telo
- če se vsi deli gibajo vzp., lahko njegovo gibanje opišemo kot gibanje točk. telesa

Kinematika

- premik

$$\vec{r} - \vec{r}' = \Delta \vec{r}$$

Δ - diferenca / razlika (konec - začetek)

d - diferencial (zelo majhna razlika)

- pot (dolžina tira)

$$s = \int_{\vec{r}'}^{\vec{r}} |\mathrm{d}\vec{r}| = \int_{r'}^r \mathrm{d}s \quad [\text{m}]$$

$$\mathrm{d}s^2 = \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2$$

- čas

časovni interval:

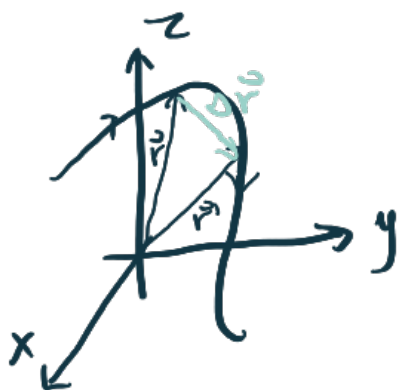
$$t - t' = \Delta t$$

- 1D:

$$s = \int_{x'}^x |\mathrm{d}x|$$

- 3D:

krivuljni integral (težko)



- Hitrost

- povprečna hitrost

$$\vec{v} = \frac{\Delta \vec{r}}{\Delta t}$$

(smer sekante na tir)

- trenutna hitrost (časovni odvod radij vektorja po času)

povp. $\rightarrow$ tren.

interval $\rightarrow$ trenutek

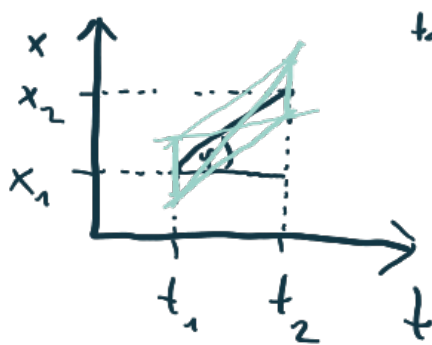
diferenka $\rightarrow$ diferencial (limitni proces)

$$\mathrm{d}t = \lim_{\Delta t \rightarrow 0} \Delta t$$

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \bar{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$

$$\vec{v} = (v_x, v_y, v_z) = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right)$$

(smer tangente na tir)



$$\tan \varphi = \vec{v}$$

problem:
zaradi napake ne moremo izmeriti
trenutne hitrosti

● - napaka

$$\vec{v} = \frac{d\vec{r}}{dt} \Rightarrow \int_{\vec{r}_1}^{\vec{r}} d\vec{r} = \int_{t'}^t \vec{v} dt$$

$$\vec{r} = \vec{r}' + \int_{t'}^t \vec{v} dt$$

- Pospešek

- povprečni

$$\bar{a} = \frac{\Delta \vec{v}}{\Delta t}$$

- trenutni

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \bar{a} = \frac{d\vec{v}}{dt}$$

$$\vec{a} = \left(\frac{dv_x}{dt}, \frac{dv_y}{dt}, \frac{dv_z}{dt} \right) = \left(\frac{d^2x}{dt^2}, \frac{d^2y}{dt^2}, \frac{d^2z}{dt^2} \right)$$

- usmerjenost:

1. $\vec{a} \parallel \vec{v}$ - spreminja velikost hitrosti

$$d\vec{v} = \vec{a} dt \parallel \vec{v} \Rightarrow |\vec{v}| = |\vec{v}'| + |\vec{a}| dt$$

2. $\vec{a} \perp \vec{v}$ - spreminja smer hitrosti



$$\sin d\varphi = \frac{|\vec{a}| dt}{|\vec{v}|} \Rightarrow d\varphi = \frac{a}{v} dt$$

$$|\vec{v}| = \sqrt{|\vec{v}'|^2 + |\vec{a}|^2 dt^2} \approx |\vec{v}'| \left(1 + \frac{1}{2} \left(\frac{|\vec{a}|}{|\vec{v}'|} \right)^2 dt^2 \right) \approx |\vec{v}'|$$

dt je zelo majhen,
 dt^2 je še manjši,
 zato lahko zanemarimo

Poskus: vozček

$$\Delta x = 10 \text{ cm} \pm 0,1 \text{ mm}$$

$$\begin{array}{l} \Delta x \\ \Delta x \\ \Delta x \end{array} \left[\begin{array}{l} \Delta t_1 \rightarrow \vec{v}_1 \\ \Delta t_2 \rightarrow \vec{v}_2 \\ \Delta t_3 \rightarrow \vec{v}_3 \end{array} \right] \quad "a" = \frac{\vec{v}_2 - \vec{v}_1}{\frac{\Delta t_1}{2} + \frac{\Delta t_2}{2}}$$

$$\Delta t = \dots \pm 1 \text{ ms}$$

Prosti pad

(enakomerno posp. gib. v 1D)

$$\begin{array}{c} y \\ \uparrow \\ 0 \end{array} \quad a_y = -g = \frac{dv_y}{dt}$$

$$\int_{v_y'}^{v_y} dv_y = \int_{t'}^t -g dt$$

$$v_y - v_y' = -g(t - t')$$

zač. čas

$$t' = 0: v_y = v_y' - gt \quad (\text{če je } t' = 0)$$

$$v_y = \frac{dy}{dt} \rightarrow \int_{y'}^y dy = \int_{t'=0}^t v_y dt = \int_0^t (v_y' - gt) dt$$

$$y = y' + v_y' t - \frac{g t^2}{2}$$

↓

$$t = \frac{-v_y' \pm \sqrt{v_y'^2 - 2g(y-y')}}{-g}$$

$$v_y = \sqrt{v_y'^2 - 2g(y-y')}$$

$$\begin{aligned} v_y &= v_y' - gt \\ -gt &= v_y - v_y' \\ t &= \frac{v_y - v_y'}{-g} \end{aligned}$$

hitreje:

$$a = \frac{dv}{dt} = \frac{dv}{dy} \cdot \frac{dy}{dt} = v \cdot \frac{dv}{dy}$$

$$a dy = v dv$$

$$\int_{y'}^y a dy = \int_{v'}^v v dv$$

$$ay - ay' = \frac{v^2}{2} - \frac{v'^2}{2}$$

$$2a(y-y') = v^2 - v'^2$$

$$v^2 = v'^2 + 2a(y-y')$$

- poševni met

y ↑

(gibanje v 2D)

- enakomerno v x
- enakomerno posp. v y
- gibanji neodvisni

$$\ddot{a}_x = 0 \quad \ddot{a}_y = -g \quad |\vec{v}| = v_0$$



$$V_x = V'_x = v_0 \cos \varphi$$

$$x = x' + v_0 \cos \varphi \cdot t$$

$$V_y = V'_y - gt = v_0 \sin \varphi - gt$$

$$y = y' + v_0 \sin \varphi t - \frac{gt^2}{2}$$

$$\text{tir: } y - y' = \tan \varphi (x - x') - \frac{g}{2v^2 \cos^2 \varphi} (x - x')^2$$

$$\text{domet: } y = y' \rightarrow t_1 = 0 \quad t_2 = \frac{2v_0 \sin \varphi}{g}$$

2x čas do najv. točke

$$d = x - x' = \frac{v^2}{g} 2 \sin \varphi \cos \varphi = \frac{v^2}{g} \sin 2\varphi; \text{ max za } \underline{\underline{\varphi = \frac{\pi}{4}}}$$

$$\varphi$$
$$x - x' = v_0 \cos \varphi t = v_0 \cos \varphi \left(\frac{2v_0 \sin \varphi}{g} \right)$$

- **Enakomerno kroženje**

(gibanje v 2D)

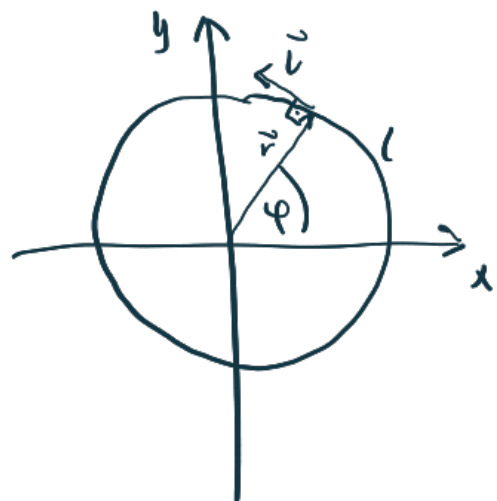
- $\varphi(t)$ - kat
- polarne koordinate
- $r = \text{konst.}$

$$\Delta \varphi = \varphi - \varphi' - \text{zasuk}$$

ločna mera: radian $\varphi = \frac{l}{r} [\text{rd}]$

katna hitrost:

$$\omega = \frac{d\varphi}{dt} [\text{s}^{-1}]$$



frekvencia (obratů čas. eh.)

$$\nu = \frac{1}{t_0} [s^{-1} = Hz]$$

t_0 - obdobi čas

obrat (zasuk 2π):

$$\omega = \frac{2\pi}{t_0} = 2\pi\nu$$

$\vec{v}$ - smer tangente $d\vec{r}$

$|\vec{v}| = r \frac{d\varphi}{dt}$

$$v = \left| \frac{d\vec{r}}{dt} \right| = r \frac{d\varphi}{dt} = r\omega$$

hitrost spreminja smer! $\Rightarrow$ posp. gibanje



$$|\Delta \vec{v}| = \lim_{\Delta t \rightarrow 0} \frac{|\Delta \vec{v}|}{\Delta t} = 2v \lim_{\Delta t, \Delta \varphi \rightarrow 0} \frac{\sin \frac{\Delta \varphi}{2}}{\frac{\Delta \varphi}{2}}$$

$$a_c = v\omega = \omega^2 r = \frac{v^2}{r}$$

Smer: $\vec{a}_c \parallel \Delta \vec{v}$ (centripetalni/radialni)

- neenakomerno kroženje
(gibanje v 2D-ravnini)

$\varphi = \varphi(t)$ - kot

$\Delta \varphi = \varphi - \varphi'$ - zasuk

$\omega = \frac{d\varphi}{dt}$ - kotna hitrost

$\alpha = \frac{d\omega}{dt}$ - kotni pospešek

lok $l = \varphi r$

obodna hitrost $v = \omega r$

tangentni pospešek $a_t = \alpha r$

skupni pospešek

$$\vec{a} = \vec{a}_r + \vec{a}_t$$

$\vec{a}_r$ spreminja smer hitrosti $\vec{a}_t$ spreminja velikost hitrosti

• opis z vektorji: enakomerno kroženje

$$\vec{r} = r(\cos\varphi, \sin\varphi) = r(\cos\omega t, \sin\omega t)$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \omega r(-\sin\omega t, \cos\omega t)$$

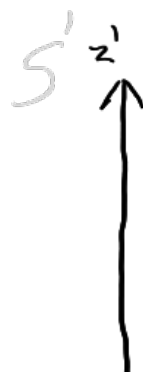
$$\vec{v} \cdot \vec{r} = 0 \Rightarrow \vec{v} \perp \vec{r}$$

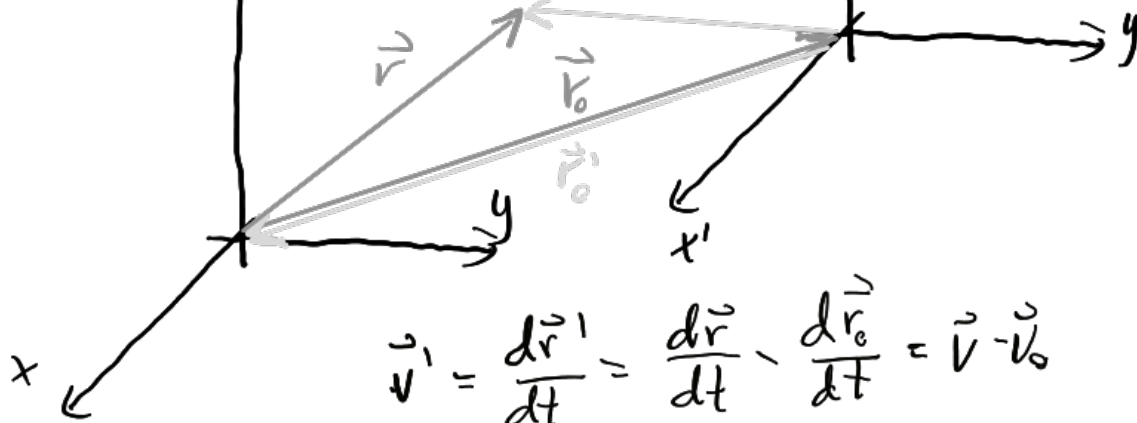
$$\vec{a} = \frac{d\vec{v}}{dt} = -\omega^2 r(\cos\omega t, \sin\omega t) =$$

$$\vec{a} = -\omega^2 \vec{r} \Rightarrow \vec{a} \parallel -\vec{r} \quad \vec{a} = \vec{a}_c$$

~ relativno gibanje

- opis gibanja iz različnih opazovalnih sistemov: S in S'
- uri tečeta enake: $t = t'$
- osi koordinatnih sistemov vzporedni:





$$\vec{v}' = \frac{d\vec{r}'}{dt} = \frac{d\vec{r}}{dt} - \frac{d\vec{r}_0}{dt} = \vec{v} - \vec{v}_0$$

$$\vec{r} = \vec{r}' + \vec{r}_0$$

$$\vec{r}' = \vec{r} - \vec{r}_0$$

$$\underbrace{\vec{r}'}_{v_{S'}} = \underbrace{\vec{r}}_{v_S} - \underbrace{\vec{r}_0}_{v_S}$$

Galilejeva transformacija: $S \rightarrow S'$

$$\vec{r} = \vec{r}' + \vec{r}_0$$

$$\underbrace{\vec{r}}_{v_S} = \underbrace{\vec{r}'}_{v_{S'}} + \underbrace{\vec{r}_0}_{v_S}$$

obratna transformacija: $S' \rightarrow S$

• sistema sta ekvivalentna $\rightarrow$ načelo relativnosti

• oba sistema sta pravilna, ampak na drugačen način merita

- inercialni opazovalni sistemi se gibljejo premo in enakomerno drug proti drugemu
- Zemlja ni zares in. op. sist. (vetrovi, tokovi)
- povzadi lahko neinercialnost na Zemlji zanemarimo (idealizacija)

- v inercialnih op. sist.: $a_0 = 0 \Rightarrow \boxed{a' = \vec{a} - \vec{a}_0 = \vec{a}}$

pospeški so v vseh in. op. sist. enaki

• primer:

Pepe in Lojze stojita vsak pred svojo hišo, Lojze vidi ptico in pokliče Pepeta in mu pove, kje je ptica. Če Pepe pogleda v smer, ki jo je povedal Lojze, ne bo videl ptice. Da bo vedel, kam mora gledati, mora vedeti,

kje je boljše in prilagoditi smer opazovanja glede na to.

Sile

-Dinamika



1. Newtonov zakon

mirovanje $\leftrightarrow$ gibanje z $v = \text{konst.}$

$\vec{v} = 0 \rightarrow \vec{v} = \text{konst.}$ - Galilejeva transformacija

(opis iz drugega inercialnega op. sistema)

Če je vsota učinkov okolice(sil) enaka nič, telo miruje ali se giblje premo in enakomerno

N.Z. veljajo le v inercialnih op. sist.

- ne veljajo upr. na vrtiljaku: (če damo kroglico gor, jo bo odneslo ven)
- ni sil $\rightarrow$ gibanje pospešeno
- mirovanje $\rightarrow$ sila (centripetalna)
- posledica sistema, ne okolice!

sila - učinek okolice/teles v okolici

definirana s pospeškom na standard za maso - sila 1N da prakilogramu pospešek $1 \frac{\text{m}}{\text{s}^2}$

$$\vec{F} \left[\text{N} = \frac{\text{kgm}}{\text{s}^2} \right]$$

masa - lastnost teles, da se upira pospeševanju

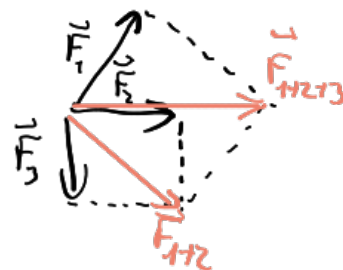
vztrajnost (inercija)

Sila 1 N: - pra-kg (m_0) $\rightarrow a_0 \approx 1 \frac{m}{s^2}$
- telo x (m_x) $\rightarrow a$

$$m_x = \frac{a_0}{a} m_0$$

pospešek vektor $\Rightarrow$ sila vektor
seštevanje sil $\Rightarrow$ seštevanje vektorjev

$$\vec{F} = \sum_i \vec{F}_i$$



masa vs. teža:

$\downarrow$
lastnost / telesa, neodv. od okolice
 $\downarrow$
vpliv okolice, sorazmeren z okolico

	Zemlja	Luna	Vesolje
masa	xxx	xxx	xxx
teža	xxx	x	0

II. Newtonov zakon

$$\sum_i \vec{F}_i = m \vec{a}$$

-vektorska enačba

$D \in \{1, 2, 3\}$ (št. dimenzij):

$$\sum_i F_{i,x} = m a_x \quad \sum_i F_{i,y} = m a_y \quad \sum_i F_{i,z} = m a_z$$

$$\sum_i \vec{F}_i = 0 \Leftrightarrow a=0 \Rightarrow \underline{\text{II. N. Z. vsebuje I. N. Z.}}$$

I. N. Z. implicira obstoj in. op. sist. (zato ga obdržimo kljub temu, da je vključen že v II. N. Z.):

Če je vsota sil na telo enaka 0, je mogoče najti op. sist.,
v katerem se telo giblje prema in enakomerno

II. N.Z. je osnova dinamike:

lahko ga razumemo kot vzročno zvezo:

$$\sum_i \vec{F}_i = m\vec{a}$$

vzrok(sila) $\rightarrow$ posledica(gibanje)

Zakoni o silah

- sile - vpliv okolice
- delovanje konkretnih teles iz okolice
- delovanje:
 - stik: podlaga, vrstica, ...
 - na daljavo: teža, el. sila
- ni „mističnih“ sil v II. N.Z. v inerc. op. sist.

- teža - privlak teles z veliko maso
 - sorazmerna z maso, smer proti S

$$\vec{F}_g = m\vec{g}$$

- meritev teže - tehtnice



- lekarniška

$$F_{g1} = F_{g2}$$

$$m_1 g = m_2 g$$

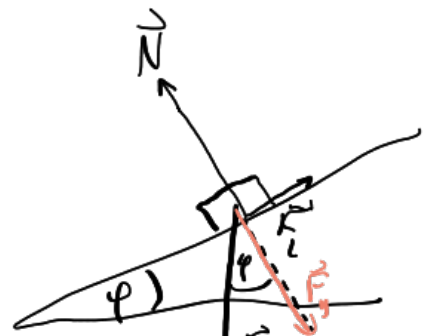
$$m_1 = m_2$$

- vzmetna

$$F_g = F_v$$

Sila podlage

- sila $\perp$ na površino (normala) $\vec{N} = -F_g \cos \varphi$
 - izniči komp. sil $\perp$ na podl.



- sila || na površino (trnjenje, lepenje)

- podlaga se upira gibanju

$$\vec{F}_l = -\vec{F}_s = -\vec{F}_g \sin \varphi$$

- približen opis

- mirovanje (lepenje)

- gibanje (trnjenje)

- F_l izniči komp. sil || s podlago

- neodvisna od vel. stične ploskve in velikosti $\vec{v}$

- zelo odvisna od stika

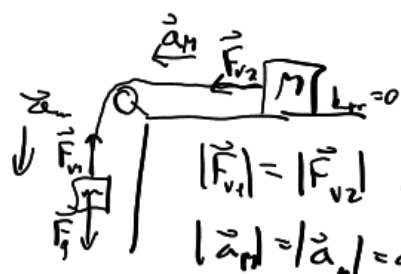
$$\begin{aligned} F_l &\leq k_l F_{\perp} \\ F_{tr} &= k_{tr} F_{\perp} \end{aligned}$$

$$\vec{F}_{tr} = -k_{tr} F_{\perp} \frac{\vec{v}}{v}$$

$$k_l \geq \tan \alpha$$

Upoštevamo samo pri mž. delu

ni žižtes; k_{tr} je malo odvisen od velikosti $\vec{v}$ ($k_l > k_{tr}$)



$$|\vec{F}_{v1}| = |\vec{F}_{v2}| \quad (\text{lahka vrv, škripec})$$

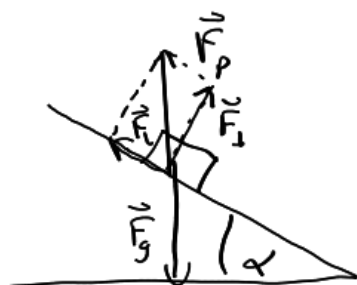
$$|\vec{a}_M| = |\vec{a}_m| = a \quad (\text{nerazt. vrv})$$

$$F_{v2} = Ma$$

$$F = mg - F_{v1} = ma$$

$$mg = Ma + ma$$

$$a = \frac{10g}{482g} \cdot 9,8 \frac{m}{s^2} \approx 0,2 \frac{m}{s^2}$$



$$\alpha = 24,6^\circ$$

$$F_l \leq k_l F_{\perp} = mg \cos \alpha \cdot k_l$$

$$F_l = mg \sin \alpha$$

$$F_{\perp} = mg \cos \alpha$$

$$k_l \geq \tan \alpha$$

$$k_l = \tan \alpha_{\max} = 0,46$$

- vrvi

- lahka vrv prenaša silo (smer in velikost)

- škripec

- lahek škripec brez trenja prenaša silo (ohrani velikost, spremenjeni smer)

- vzmet

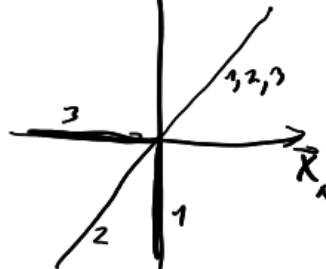
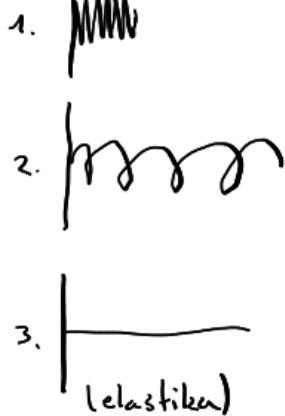
- sila nasprotna raztežku

- velikost sile sorazmerna raztežku

$$\vec{F}_{vz} = -k \vec{x} \quad \text{Hooke}$$

velja, dokler se vzmet ne deformira (se ne poruši v začetno stanje)

$$\vec{F}_{vz} \uparrow$$



III. Newtonov zakon

- zakon o vzajemnem učinku

$$\vec{F}_{BC} = -\vec{F}_{CB}$$

- če deluje prvo telo na drugo z neko silo, deluje drugo telo na prvo z nasprotno enako silo

- naravni zakon - povzetele opazovanj (velja za vse Newt. zakone)

- NI VZROČNA ZVEZA (če zamenjamo B in C, se zakon ne spremeni)

- primer: atlet pri skoku v višino: $\vec{F}_g = -\vec{F}_A$

- kakšen Zemlja pade proti atletinji?

$$m_2 \cdot a_2 = m_1 \cdot a_1$$

$$a_2 = a_1 \frac{m_1}{m_2} = g \cdot \frac{60 \text{ kg}}{6 \cdot 10^{24} \text{ kg}} \approx 10^{-23} g = 10^{-22} \frac{\text{m}}{\text{s}^2}$$

nemerljivo malo,
preveč sumav (npr. nekdo
skoči na drugi strani
Zemlje)

Sile pri kroženju

- telo enakomerno kroži $\Rightarrow \vec{a}_c \Rightarrow \vec{F}_c = m \vec{a}_c$

- če sile ni $\Rightarrow$ smer tangente na tir ($\vec{v}$)

- Luna okoli Zemlje $F_c \approx F_g$

- kroglica na štriku $F_c \approx F_v$

- avto na ovinku $F_o \approx F_l$

- lahko povečamo z:

- nagnjen ovinek (vpliv F_l na F_c se zmanjša, poveča se vpliv N)

- srk zraka (aerodinamika avta)

Neinercialni op. sistemi

$$\vec{a}_{\text{nein}} = \vec{a}_{\text{in}} - \vec{a}_o$$

pospešek notn. op. sist., izmerjen v in. op. sist.

$$\vec{F}_{\text{inert}} = m \vec{a}_{\text{in}} \Rightarrow \vec{F} = m \vec{a}_{\text{nein}} \\ \vec{F} = m \vec{a}_{\text{in}} \quad \vec{F} = m \cdot \vec{a}_{\text{nein}} = m(\vec{a}_{\text{in}} - \vec{a}_0)$$

Stvari, ki niso del sistema:

a) $m_{\text{nein}} = m_{\text{in}} = m \checkmark$

b) $\vec{F}_{\text{nein}} = \vec{F}_{\text{in}} = \vec{F} \checkmark$

$$\vec{a}_0 = 0$$

Newtonovi zakoni veljajo samo v in. op. sist. (k. je $\vec{a}_0 = 0$)

$$\vec{F} - m\vec{a}_0 = m\vec{a}_{\text{nein}}$$

SYSTEMSKE
SILE

$\vec{F}_s = -m\vec{a}_s \leftarrow$ Na Zemlji ne ločimo F_g od F_s

$$\vec{F} + \vec{F}_s = m\vec{a}_{\text{nein}}$$

upliv okoličice

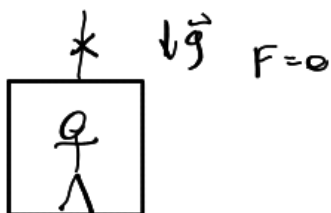
posledica tega, da
nismo v in. op. sist.

F_s deluje na daljavo, deluje na maso

a) dvigalo miruje

$$\vec{F}_p + \vec{F}_g = 0$$

b) prerežemo žico



Kroženje v nein. sist.



(gremo v kompl. ravn.)
zaradi polarnega zapisa

$$\vec{r} = z$$

$$\vec{r}_e = z_e$$

$$z = z_e e^{i\omega t}$$

zaradi prehoda iz in. v nein. sist.

$$v_c = \frac{dz}{dt} = \frac{dz_e}{dt} e^{i\omega t} + i\omega z_e e^{i\omega t}$$

rotacija z_e
za $90^\circ \rightarrow e^{i\frac{\pi}{2}}$

$$\vec{v}_e = e^{i\omega t} \left(\frac{d\vec{z}_e}{dt} + i\omega \vec{z}_e \right) \text{ za poljuben vektor}$$

(vektarji so isti v in in nein op. sist.)

$$\vec{v} = \vec{\omega} \times \vec{r}$$

$$\boxed{\frac{d}{dt_i} = \frac{d}{dt_e} + \vec{\omega} \times \vec{r}}$$

$$\vec{a}_i = \frac{d\vec{v}_i}{dt_i} = e^{i\omega t} \frac{d\vec{v}_e}{dt} + \overbrace{\vec{\omega} \times \vec{v}_e}^{=i\omega \vec{v}_e} =$$

$$= e^{i\omega t} \left(\frac{d^2 \vec{z}_e}{dt^2} + i\omega \vec{v}_e + i\omega \vec{v}_e - \omega^2 \vec{z}_e \right)$$

$$2 \vec{\omega} \times \vec{v} = \vec{a}_{\text{Coriolis}}$$

Coriolisov pospešek

$$\vec{F}_{\text{Coriolis}} = -m \vec{a}_{\text{Coriolis}} = -2m \vec{\omega} \times \vec{v}$$

oblak, kjer se nekaj zgodi

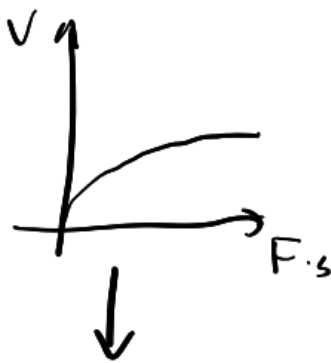


$$\vec{F} = m \vec{a} = m \frac{d\vec{v}}{dt}$$

$$\int \vec{F} dt \leftarrow \vec{F}(t)$$

$$\downarrow$$

$$\underline{\int \vec{F} \cdot d\vec{s}} = \int \vec{F}(t) \cdot \vec{v}(t) dt = \underline{\int P(t) dt = A}$$



= mokre sanje usazga fizikov

$$\downarrow$$

$$A \rightarrow W_k = \frac{m}{2} v^2$$

algebraično:

$$\vec{v} \rightarrow d\vec{v}$$

$$\vec{F} = m\vec{a} = m \frac{d\vec{v}}{dt}$$

$$\vec{F} \cdot d\vec{s} = m \frac{d\vec{v}}{dt} \cdot d\vec{s} = m \vec{v} d\vec{v}$$

$$\int \vec{F} \cdot d\vec{s} = \int m \vec{v} d\vec{v}$$

$$A = m \frac{v^2}{2} \Big|_{v_0}^v = \Delta W_k$$

$$A = 0$$

- sila pravokotna na tir gibanja

• F_c (centripetalna)

• F_m (magnetna)

- ni premika

• Atrpeja

$$A_{tr} = \int \vec{F}_{tr} \cdot d\vec{r} =$$

$$= -k_{tr} F_p \int \frac{\vec{v}}{v} \cdot \vec{v} dt =$$

$$= -k_{tr} F_p \int \frac{v^2}{v} dt =$$

$$= -k_{tr} F_p \int v dt < 0$$

Disipativne sile (delo teh sil je vedno < 0)

↓
• upor
• trenje

↓
povzročajo
izgubo W_k

Sila je konzervativna, če $A = 0$, ko se začne in konča v isti točki

↓
npr. F_g, F_{vzn}

ENERGIJA (lastnost telesu)

- kinetična (iz II. N.Z.) ← gibanje

- potencialna ← konzervativne sile

lega (glede na druga telesa)

$$A = \Delta W_k$$

$$A_k + A' = \Delta W_k$$

$$\Delta W_p = -A_k$$

- teža

- vzmet

$$A' = \Delta W_k + \Delta W_p$$

$$\text{teža: } A_g = -mg(h-h')$$

$$\Delta W_{pg} = mg(h - h')$$

$$\text{vzmet: } A_v = \int_{x'}^x \vec{F}_v \cdot d\vec{s} = \int_{x'}^x -kx dx = -\frac{k}{2}(x^2 - x'^2)$$

$$\Delta W_{p,pr} = \frac{k}{2}(x^2 - x'^2)$$

$$\Delta W_p(\vec{r}, \vec{r}') = W_p(\vec{r}) - W_p(\vec{r}')$$

referenčna lega

→ vrednost, povzadi 0 (lahko tudi ∞)



ko vzmet ni napeta, nima prožnostne energije

$$W_{pr} = \Delta W_p(x, 0)$$

$$W_p(0) = 0$$

$$W_{pr} = \frac{kx^2}{2}$$



$W_{p,g}$ nima naravne ref. lege

$$A' = \Delta W_k + \Delta W_p \quad \text{izrek o kin. in pot. energiji}$$

↑
delo vseh
sil iz okolice

$$A' = 0 \Rightarrow \text{ohranitev } \Delta W_p \text{ in } \Delta W_k$$

$$W_k + W_p = \text{konst.}$$

Energijski zakon:

- v izoliranem sistemu se energija ohranja.
P
ni vpliva okolice

Moč (hitrost dela)

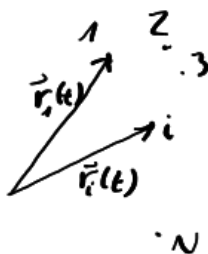
$$P = \frac{dA}{dt} \quad \left[W = \frac{J}{s} \right] \quad k W_h = 3,6 MJ$$

$$P = \frac{\vec{F} \cdot (d\vec{s})}{dt} = \vec{F} \cdot \vec{v}$$

$P = \frac{dW}{dt}$ - hitrost pretvarjanja energij

Sile med telesi

$\sum \vec{F}_{ji}$ - notranje sile
 $\uparrow$
 sila i na j



$\boxed{\vec{F}_{ij} = -\vec{F}_{ji}}$ III. N.Z.

$\sum \vec{F}_{i,z}$ - zunanje sile

Ali je v sist. točk. teles točka, ki se giba kot točk. telo?

$$\left. \begin{aligned} m_1 \vec{a}_1 &= \vec{F}_{1,zun} + \sum_{j=2}^N \vec{F}_{1j} \\ m_i \vec{a}_i &= \vec{F}_{i,zun} + \sum_{j=1, j \neq i}^N \vec{F}_{ij} \end{aligned} \right\} +$$

sile nastopajo v parih
 zaradi III. Newt. zakona

$$m_1 \vec{a}_1 + m_2 \vec{a}_2 + \dots + m_i \vec{a}_i + \dots + m_N \vec{a}_N = \sum_i \vec{F}_{i,zun} + 0$$

$$\frac{d^2}{dt^2} (m_1 \vec{r}_1 + m_2 \vec{r}_2 + \dots + m_i \vec{r}_i + \dots + m_N \vec{r}_N) = \vec{F}_{zun}$$

$$M \vec{r}^* = \frac{\sum m_i \vec{r}_i}{\sum m_i} = \frac{\int \vec{r} dm}{\int dm} = \frac{\int \vec{r} \rho(\vec{r}) dV}{M}$$

Težišče:

$A = \Delta W_k$
 $A_{zun} \neq A_{not}$
 $\uparrow$
 pr. prihaji
 delo opravljajo
 miške, ki so
 del sistema

$$\vec{F}_{zun} = M \vec{a}^* \quad / \quad \int d\vec{r}^*$$

$$\int \vec{F}_{zun} d\vec{r}^* = M \int \vec{a}^* d\vec{r}^* \frac{d\vec{r}^*}{dt} = \frac{M}{2} (\vec{v}^{*2} - \vec{v}^{*2}) = \Delta W_k$$

translacijska kinetična en.
 (ko se vsi deli telesa gibajo z enako hit.)

IZREK O W_k TEŽIŠČA/
 IZREK O TRANSL. W_k

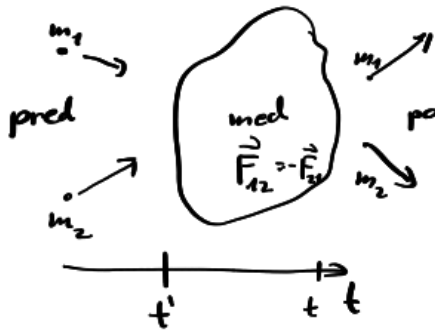
$$\sum A_{ij} = \int \vec{F}_{ij} d\vec{r}_i + \int \vec{F}_{ji} d\vec{r}_j = \int \vec{F}_{ij} (d\vec{r}_i - d\vec{r}_j) = \int \vec{F}_{ij} d(\vec{r}_i - \vec{r}_j)$$

III. N.Z. če $\vec{F}_{ij}$ ima smer $\vec{r}_i - \vec{r}_j \Rightarrow A=0$

↓
centralne sile

če $d(\vec{r}_i - \vec{r}_j) = 0 \Rightarrow$ telo se ne deformira
togo telo $\Rightarrow A_n = 0$

trki:



$$m_1 \ddot{a}_1 = \vec{F}_{12} / \int dt$$

$$\int m_1 d\vec{v}_1 = \int \vec{F}_{12} dt$$

$$m_1 (\vec{v}_1 - \vec{v}_1') = \int \vec{F}_{12} dt \quad \text{sumek sile}$$

Gibalna količina: $\vec{G} = m\vec{v}$ (linear momentum)

$$\vec{F} = m\ddot{a} = \frac{d\vec{G}}{dt}$$

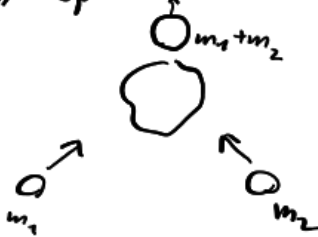
IZREK O GIBALNI KOLIČINI

$$\int \vec{F} dt = \Delta \vec{G}$$

$$\vec{G}_{sist.} = \vec{G}_1 + \vec{G}_2 = m_1 \vec{v}_1 + m_2 \vec{v}_2 = (m_1 + m_2) \vec{v}^*$$

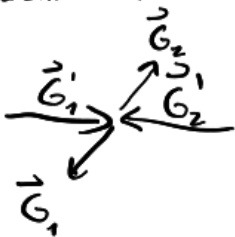
Ker F_{12} not. sila: $\Delta \vec{G}_{sist.} = 0$

(popolnoma) neprožni trk:



$$m_1 \vec{v}_1 + m_2 \vec{v}_2 = (m_1 + m_2) \vec{v}$$

prazni trk



centralni trk: vse se dogaja na premici



$$v_2' = 0 :$$

$$m_1 v_1' = m_1 v_1 + m_2 v_2$$

$$m_1 (v_1' - v_1) = m_2 v_2$$

$$m_1 v_1'^2 = m_1 v_1^2 + m_2 v_2^2$$

$$m_1 (v_1' + v_1)(v_1' - v_1) = m_2 v_2^2$$

$$v_1' + v_1 = v_2$$

$$m_1 (v_1' - v_1) = m_2 (v_1' + v_1)$$

$$(m_1 - m_2) v_1' = v_1 (m_1 + m_2)$$

$$v_1 = v_1' \frac{m_1 - m_2}{m_1 + m_2}$$

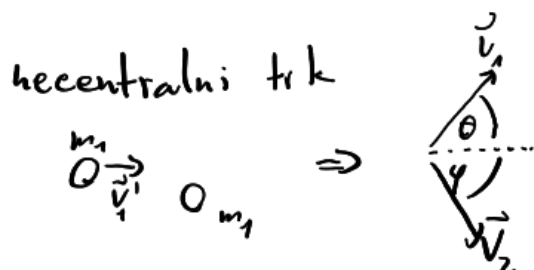
$$v_2 = v_1' \left(\frac{m_1 + m_2 + m_1 - m_2}{m_1 + m_2} \right)$$

$$v_2 = v_1' \left(\frac{2m_1}{m_1 + m_2} \right)$$

$$m_1 = m_2 \quad v_1 = 0 \quad v_2 = v_1'$$

$$m_1 \ll m_2 \quad v_1 \sim v_1' \quad v_2 \sim 0$$

$$m_1 \gg m_2 \quad v_1 \sim v_1' \quad v_2 \sim 2v_1'$$

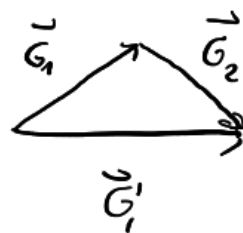


$\varphi: [0, \frac{\pi}{2}]$
pri cent. trku
ko se telesi rahlo oplazita

$$m v_1' = m v_1 \cos \theta + m v_2 \cos \varphi$$

$$m v_1 \sin \theta = m v_2 \sin \varphi$$

$$m v_1'^2 = m v_1^2 + m v_2^2$$



$$\vec{G}_1 + \vec{G}_2 = \vec{G}_1' / 2$$

$$m_1^2 v_1^2 + m_2^2 v_2^2 + 2 \vec{G}_1 \cdot \vec{G}_2 = m_1^2 v_1'^2$$

če $m_1 = m_2$:

$$\vec{G}_1 \cdot \vec{G}_2 = 0$$

$$\Rightarrow \theta + \varphi = \frac{\pi}{2}$$

Gravitacija

$$\vec{F} = -G \frac{m_1 m_2}{r^2}$$

$$G = 6,67 \cdot 10^{-11} \frac{\text{Nm}^2}{\text{kg}^2} \quad (\text{gravitacijska konstanta})$$

$$g(h) = G \frac{M_2}{(R_2 + h)^2} \cdot \frac{R_2^2}{R_2^2} = g_0 \left(\frac{R}{R + h} \right)^2 = g_0 \frac{1}{\left(1 + \frac{h}{R_2}\right)^2} = g_0 \frac{1}{1 + \frac{2h}{R_2} + \frac{h^2}{R_2^2}}$$

$$h \ll R_2 \rightarrow g(h) = g_0 \left(1 - \frac{2h}{R_2}\right)$$

$$\Delta W_p = -A_g = - \int_{\vec{r}_2}^{\vec{r}_k} \vec{F}_g d\vec{r} = - \left(-GmM \int_{r_2}^{r_k} \frac{\vec{r} d\vec{r}}{r^3} \right) = GmM \int_{r_2}^{r_k} \frac{dr}{r^2} = -GmM \left(\frac{1}{r_k} - \frac{1}{r_2} \right)$$

$$\vec{r} d\vec{r} = r dr$$

$$W_p(\infty) = 0$$



$$W_{pg} = -\frac{GmM}{r}$$



$$W = W_k + W_{pg} = \text{konst.}$$

- kometi $\leftarrow W > 0 \dots$ prost sistem
 - če sta daleč stran,
 se še vedno gibata - hiperbola

- planeti $\leftarrow W < 0 \dots$ vezan sistem - elipsa
 - kometi

$W = 0 \dots$ parabola

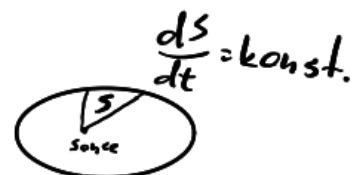
Kepplerjevi zakoni

1. Tiri planetov so elipse, sonce v enem od gorišč

2. Ploščinska hitrost je konst.

3. $\frac{t_o^2}{a^3} = \text{konst.}$ za vse planete

p
velike polosi



za fizike:

1. Planeti krožijo
2. $\omega = \text{konst.}$
3. $\frac{t_0^2}{r^3} = \text{konst.}$ za vse planete

$$F_c = F_g$$

$$m_p \omega^2 r = G \frac{m_p m_s}{r^2}$$

$$\omega = 2\pi \frac{1}{t_0}$$

$$\frac{4\pi^2}{Gm_s} = \frac{t_0^2}{r^3}$$

↓
konst.

$$F_c = F_g$$

$$\frac{1}{2} \cdot m_p \frac{v^2}{r} = G \frac{m_p m_s}{r^2} \cdot \frac{1}{2}$$

$$W_k = -\frac{1}{2} W_p$$

Kozmične hitrosti

① kroženje okoli Zemlje v nizki orbiti

$$\frac{v^2}{R_z} = g_0 \rightarrow v_I = \sqrt{g_0 R_z} = 8 \frac{\text{km}}{\text{s}} \quad \leftarrow \text{1. kozmična hitrost}$$

② Ubežna hitrost za Zemljo

$$W_k + W_p = 0$$

$$\frac{1}{2} m v_{II}^2 = G \frac{m M_z}{R_z}$$

$$v_{II} = \sqrt{\frac{2GM_z R_z}{R_z R_z}} = \sqrt{2g_0 R_z} = v_I \sqrt{2} = 11 \frac{\text{km}}{\text{s}} \quad \leftarrow \text{2. kozmična hitrost}$$

③ Ubežna hitrost za Sončni sistem

$$d = 1,5 \cdot 10^{11} \text{ m}$$

$$W_k + W_p = 0$$

$$\frac{1}{2} m v^2 = G \frac{m M_s}{d}$$

$$v = \frac{2\pi d}{1 \text{ leto}} = \frac{2\pi \cdot 1,5 \cdot 10^{11} \text{ m}}{1 \cdot 10^7 \text{ s}} = 30 \frac{\text{km}}{\text{s}}$$

$$M_s = \frac{d v^2}{G} = \frac{1,5 \cdot 10^{11} \text{ m} \cdot 9 \cdot 10^8 \text{ km}^2}{6,67 \cdot 10^{-11} \text{ N m}^2 \text{ kg}^{-2}} = 2 \cdot 10^{30} \text{ kg}$$

$$\frac{m v_{\text{II}}^2}{2} = G \frac{M_1 M_2}{d}$$

$$6,67 \cdot 10^{-11} \text{ N m}^2/\text{kg}^2$$

$$2 \cdot 10^3 \text{ kg}$$

$$v_{\text{II}}' = \sqrt{\frac{2 M_2 G}{d}} = 4,2 \cdot 10^4 \frac{\text{m}}{\text{s}} = 42 \frac{\text{km}}{\text{s}}$$

$$v_{\text{II}} = \sqrt{(v_{\text{II}}' - v_z)^2 + v_{\text{II}}^2} = \underline{\underline{16,6 \frac{\text{km}}{\text{s}}}} \leftarrow \text{3. kozmična hitrost}$$

če izstrelimo ob pravih trenutkih, imamo $30 \frac{\text{km}}{\text{s}}$ iz os gibanja Zemlje okoli Sonca

Vrtenje

- preprosto:

- fiksna os
- togo telo



$d\varphi$ - zasuk

- enak po celotnem telesu

$\Downarrow$

ω enaka po celotnem telesu

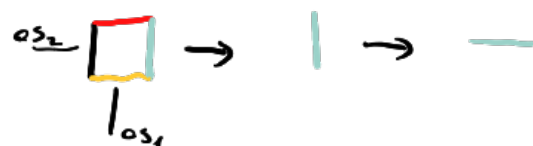
$\Downarrow$

α - 11-

$\vec{\omega}, \vec{\alpha}$ imajo smer osi $\Rightarrow$ vrtenje okoli fiksne osi
je 1-D

pozitivna smer vrtenja - v nasprotni smeri vrtenja urinega kazalca

zasuk v 3D NI vektor ($d\vec{\varphi}_1 + d\vec{\varphi}_2 \neq d\vec{\varphi}_2 + d\vec{\varphi}_1$)



$d\vec{\varphi}$ v 3D je vektor!

$\vec{\omega}, \vec{\alpha}$ - 11-

$$W = \frac{1}{2} I \omega^2 = \frac{1}{2} (I_1 \omega_1^2 + I_2 \omega_2^2 + I_3 \omega_3^2) = \frac{1}{2} \omega^2 I$$

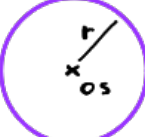
$$J_k = \frac{1}{2} \int V^2 dm = \frac{1}{2} \int dm \omega^2 r^2 = \frac{1}{2} \omega^2 \int dm r^2 = \frac{1}{2} \omega^2 J$$

vztrajnostni moment

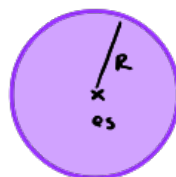
točka

$\dots r \cdot x_{os}$ $J = mr^2$

obroč,
cevi

 $J = Mr^2$

plošča,
valj



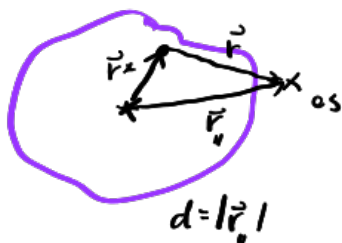
razbijemo na več obročev/cevi

$$dm = \rho dV = \rho 2\pi r dr h$$

$$J = \int dm r^2 = \rho 2\pi h \int_0^R r^3 dr = \rho 2\pi h \frac{R^4}{4} = \frac{1}{2} MR^2$$

$$M = \rho \cdot V = \rho \pi R^2 h$$

telu, kjer je
os premaknjena
za d iz
težišča



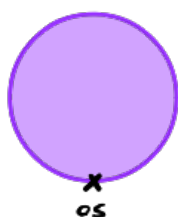
$$\vec{r} = (\vec{r}_* - \vec{r}_{||})$$

$$J_{||} = \int r^2 dm = \int (\vec{r}_* - \vec{r}_{||})^2 dm = \int dm r_*^2 + \int dm r_{||}^2 - 2 \int dm \vec{r}_* \cdot \vec{r}_{||}$$

$$J_{||} = J^* + md^2$$

← Steinerjev izrek

(SAMO ZA VZPOREDNE PREG MIKE OSI)

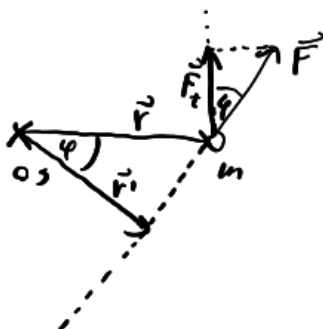


$$J = \frac{1}{2} mr^2 + mr^2 = \frac{3}{2} mr^2$$

Navor

$$M = F_t r = r F \sin \varphi$$

$$\vec{M} = \vec{r} \times \vec{F}$$



$$F_t = ma_t$$

$$M = r \cdot F_t = m a_t \cdot r = J \cdot \alpha \quad \text{— za togo telo}$$

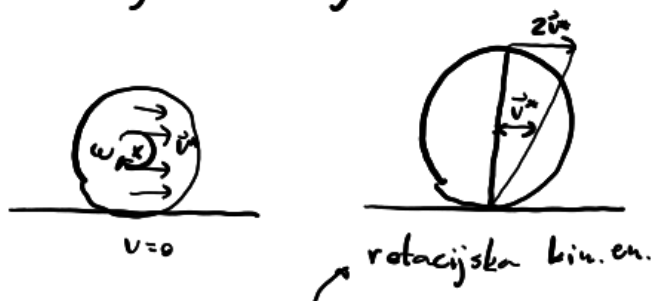
gibanje v 1D:

premo	vrtenje
x	y
v_x	ω

$$dA = \vec{r} \cdot d\vec{s} = r_t dl = r_t \cdot r \cdot d\varphi = r l d\varphi$$

$$P = \frac{dA}{dt} = M \frac{d\varphi}{dt} = M\omega$$

Kotaljenje
(translacija + rotacija)



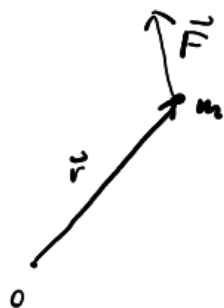
$$W_k = \frac{1}{2}mv^2 + \frac{1}{2}J\omega^2$$

npr. za kroglo $\frac{1}{2} \cdot \frac{2}{5} m R^2 \cdot \frac{v^2}{R^2}$

Vrtenje v 3D

$\odot \rightarrow$ osi šče (točka, okrog katere se vrti telo) [0]
(izhodišče)

- ročica $[\vec{r}]$



moment $\vec{M} = \vec{r} \times \vec{F}$

$$\vec{F} = \frac{d\vec{G}}{dt}$$

$$\vec{r} \times \vec{F} = \frac{d}{dt} (\vec{r} \times \vec{G})$$

$$\vec{r} \times \vec{G} = \vec{L}$$

vrtilna količina

$$\vec{M} = \frac{d\vec{L}}{dt}$$

Sistem teles

a_x	$\alpha (+a_r)$
m	J
$\frac{mv^2}{2}$	$J\frac{\omega^2}{2}$
$F=ma$	$M=J\alpha$
$dA=Fds$	$dA=Md\varphi$
$P=Fv$	$P=M\omega$

$$\vec{L} = \sum_i m_i \vec{r}_i \times \vec{v}_i$$

$$\vec{L} = \int dm \vec{r} \times \vec{v}$$

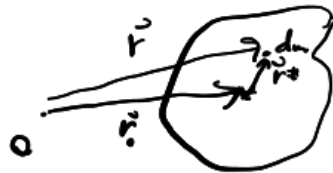

$\sim m_i$ novoro kot. sil

$$\sum \vec{M}_{\text{zun}} = \frac{d\vec{L}}{dt}$$

velja za inercialno osišče

če je osišče v težišču:

$$\vec{M}^* = \frac{d\vec{L}^*}{dt}$$



$$\sum \vec{M}_{\text{zun}} = \frac{d\vec{L}}{dt}$$

$$\sum \vec{r} \times \vec{F}_{\text{zun}} =$$

$$= \sum \vec{r}_0 \times \vec{F}_{\text{zun}} + \sum \vec{r}^* \times \vec{F}_{\text{zun}} =$$

$$= \vec{r}_0 \times \sum \vec{F}_{\text{zun}} + \vec{M}_{\text{zun}}^*$$

$$\vec{L} = \int \vec{r} \times d\vec{G}$$

$$= \int (\vec{r}^* + \vec{r}_0) dm \times (\vec{v}_0 + \vec{v}^*) =$$

$$= \int dm [\cancel{\vec{r}^* \times \vec{v}_0} + \vec{r}_0 \times \vec{v}_0 + \vec{r}^* \times \vec{v}^* + \cancel{\vec{r}_0 \times \vec{v}^*}] =$$

$$= \vec{r}_0 \times \vec{G}_0 + \int dm \vec{r}^* \times \vec{v}^* =$$

$$= \vec{r}_0 \times \vec{G}_0 + \vec{L}^*$$

$$\frac{d\vec{L}}{dt} = \vec{r}_0 \times \frac{d\vec{G}_0}{dt} + \frac{d\vec{L}^*}{dt} = \vec{r}_0 \times m\vec{a}_0 + \frac{d\vec{L}^*}{dt}$$

$$\begin{aligned} &= \underbrace{\left(\int dm \vec{r}^* \right)}_{=0} \times \vec{v}_0 = 0 \\ &= \vec{r}_0 \times \int dm \vec{v}^* = \\ &= \vec{r}_0 \times \underbrace{\int dm \frac{d\vec{r}^*}{dt}}_{=0} = 0 \end{aligned}$$

$$\vec{L} = \int \vec{r} \times d\vec{G} = \int dm \vec{r} \times \vec{v} = \int dm \underbrace{\vec{r} \times (\vec{\omega} \times \vec{r})}_{\lambda \vec{\omega} + \mu \vec{v}}$$

$$\vec{L} = J\vec{\omega}?$$

$$J = \rho^2 dm$$

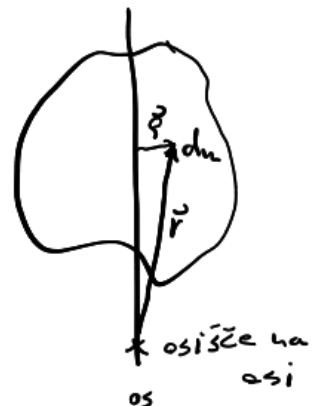
$$(\vec{r} \cdot \vec{r})\vec{\omega} - (\vec{r} \cdot \vec{\omega})\vec{r}$$

$$r^2 \vec{\omega} - r_{\omega}^2 \vec{\omega} = \rho^2 \vec{\omega}$$

$$(-r_{\omega} \vec{\omega} \vec{r})$$

$$\vec{L}_{\omega} = \int dm \rho^2 \vec{\omega} = J_{10} \vec{\omega}$$

$$\vec{L} = \vec{L}_{\omega} + \vec{L}_c \Rightarrow \vec{L} \neq \vec{\omega}$$



$$J_{\text{ob}} \cdot \vec{\omega}$$

$$\vec{\Gamma} = J \cdot \vec{\omega} \quad \text{tenzor}$$

$$\vec{\omega} = (\omega_x, \omega_y, \omega_z) \quad \vec{r} = (x, y, z)$$

$$\vec{\Gamma} = (\Gamma_x, \Gamma_y, \Gamma_z) = \int dm [(r^2 \cdot \vec{r}) \vec{\omega} - (\vec{r} \cdot \vec{\omega}) \vec{r}] =$$

$$= \int dm [(x^2 + y^2 + z^2) \vec{\omega} - (x\omega_x + y\omega_y + z\omega_z) \vec{r}] =$$

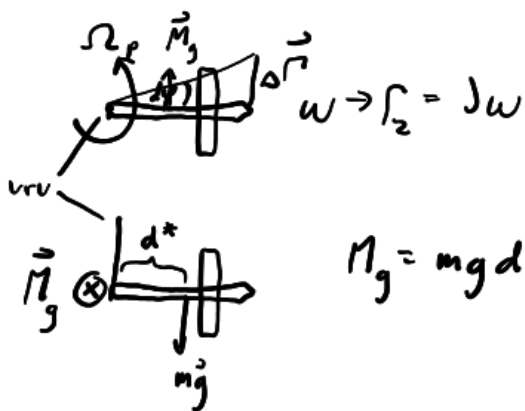
$$= \int dm [(x^2 + y^2 + z^2 - x^2 - xy - xz) \omega_x +$$

$$(x^2 + y^2 + z^2 - y^2 - xy - yz) \omega_y +$$

$$(x^2 + y^2 + z^2 - z^2 - xz - yz) \omega_z]$$

Precesija

(vrtavka)



$$M_g = mgd^*$$

$$mgd^* = M_g = \frac{d\Gamma}{dt} = \Omega_p J \omega$$

$$\boxed{\Omega_p = \frac{mgd^*}{J\omega}}$$

zaradi precesije se pri vrtavki

ω_L pretvarja v ω_P in obratno

